

GMT

Green Metrics
Technology

Team 7

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Buildable Areas for Laneway Homes

Final presentation

Green Metrics Technology Corp.

GMT

Green Metrics
Technology Corp.



Vancouver, BC



Real estate decarbonization



AI-powered building analytics

Our Partner

What They Do

GMT helps housing managers, developers, and modular manufacturers make data-driven decisions to optimize building performance, reduce carbon footprints, and simplify decarbonization investments.

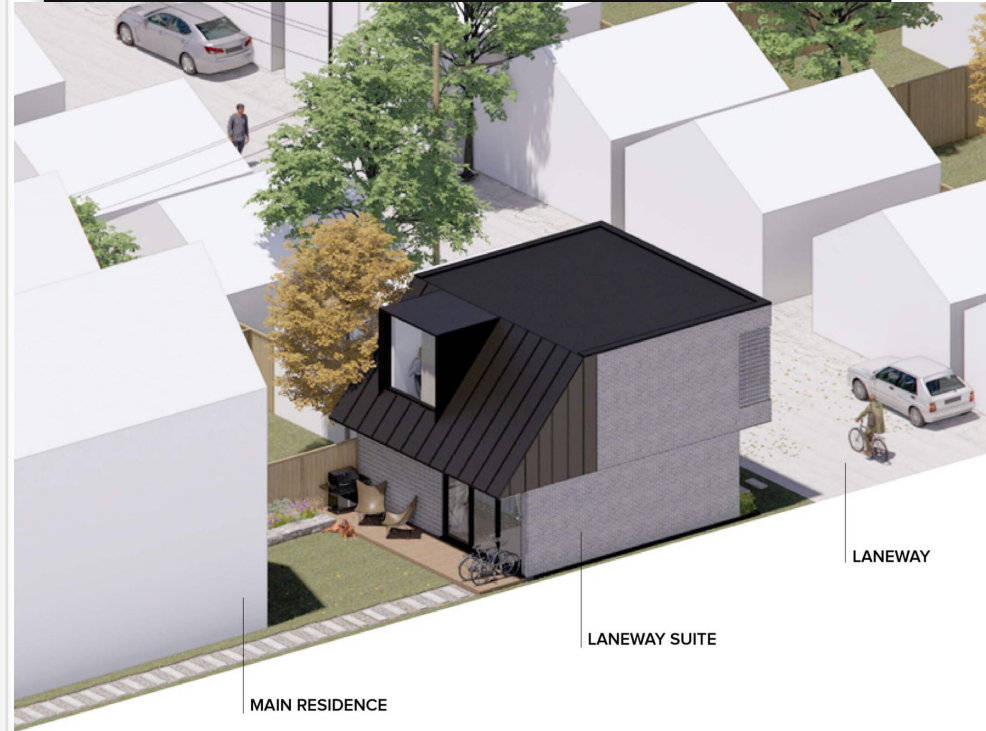
Why This Project

GMT partnered with our team to explore how AI can automate zoning bylaw analysis — turning hundreds of pages of municipal rules into structured, map-ready data that supports faster, smarter development decisions.

Project Overview

"I want to build a laneway home in my property, but..."

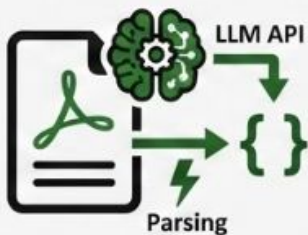
- 1 Can I build it?
- 2 How much space do I have?
- 3 Are there other restrictions?



System Architecture Overview

End-to-end pipeline: from raw PDF bylaw to buildable area output

Stage 1 Rule Extraction



PDF -> Structured JSON
Rules via LLM

Stage 2 Verification Layer



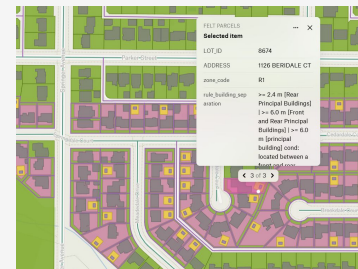
Validating and Flagging
extracted rules

Stage 3 GIS Rule Contract



Map setbacks, compute
buildable area with
provenance

Stage 4 Map Dashboard



Combine relevant rules
with GIS -> clickable of the
parcel

Stage 1: Rule Extraction

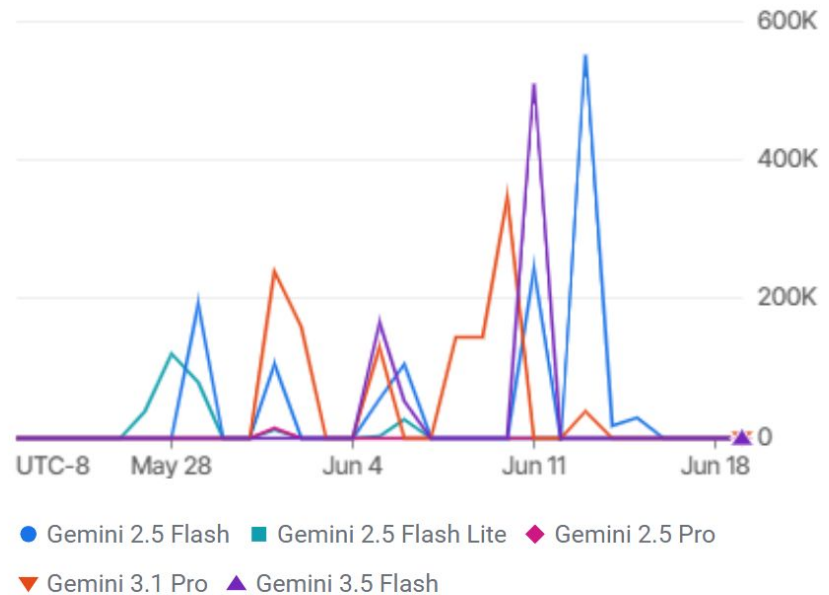
PDF Bylaw → Structured Rules via LLM Pipeline

Rule Extraction Challenges

1 Locally deployed LLM not smart enough

2 Online API costly

Output Tokens per model



Rule Extraction Challenges

	Dwelling Type			
	Rowhouse	Small-Scale Multi-Unit		
Permitted Dwelling Units (including secondary suites)	1 to 3 Units	1 to 2 Units	3 to 4 Units	5 to 6 Units Frequent Transit Network Area Only
Minimum Lot Area	-	-	4 Units Only: 281 m ²	281 m ²
Maximum Lot Area ¹	280 m ²	-	-	-
Maximum Lot Coverage				
		Lots < 567 m ² : 40%		
All Buildings	55%	Lots > 567 m ² : 30%	40%	45%
Impervious Surfaces	70%	60%	60%	70%
Maximum Height				
Front Principal Buildings				
Height	sloping roof: 10 m flat roof: 9.5 m			
Storeys (basement inclusive)	3 storeys	2.5 storeys	3 storeys	
Rear Principal Buildings				
Height	sloping roof: 7.5 m flat roof: 7.0 m			
Storeys (basement inclusive)	2 storeys			
Accessory Buildings	4.0 m 1 storey			
Minimum Lot Line Setbacks for All Buildings ^{2,3}				
Street Yard	3.0 m	front: 4.0 m flanking: 3.0 m		
Lane Yard	1.5 m			
Interior Rear Yard	3.0 m, except 1.5 m for accessory buildings			
Interior Side Yard	0 m, except 1.2 m for end unit lots	1.2 m	1.2 m	1.2 m
Minimum Separation of Buildings on the Same Lot ^{4,5}				
Between Front Principals	-	2.4 m	2.4 m	2.4 m
Between Rear Principals	-	2.4 m	2.4 m	2.4 m
Between Front & Rear Principals	6.0 m			
Between All Other Buildings	2.4 m			

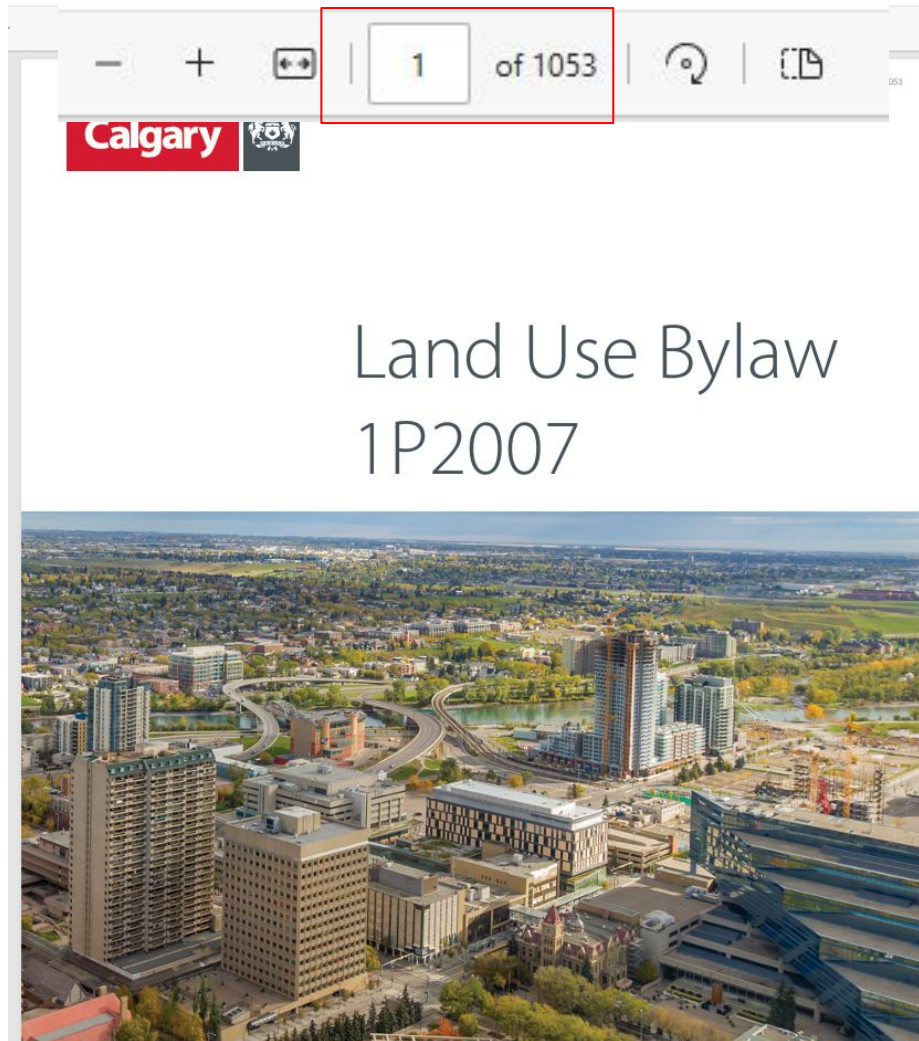


Field	Value
normalized rule key	max_lot_coverage_all_buildings
rule object	All Buildings
operator	<=
value	30
unit	percent
condition	Small-Scale Multi-Unit / 1 to 2 Units / Lots > 567 m ²
source stream	deterministic_table
page number	2
verification_status	verified
final action	ACCEPT

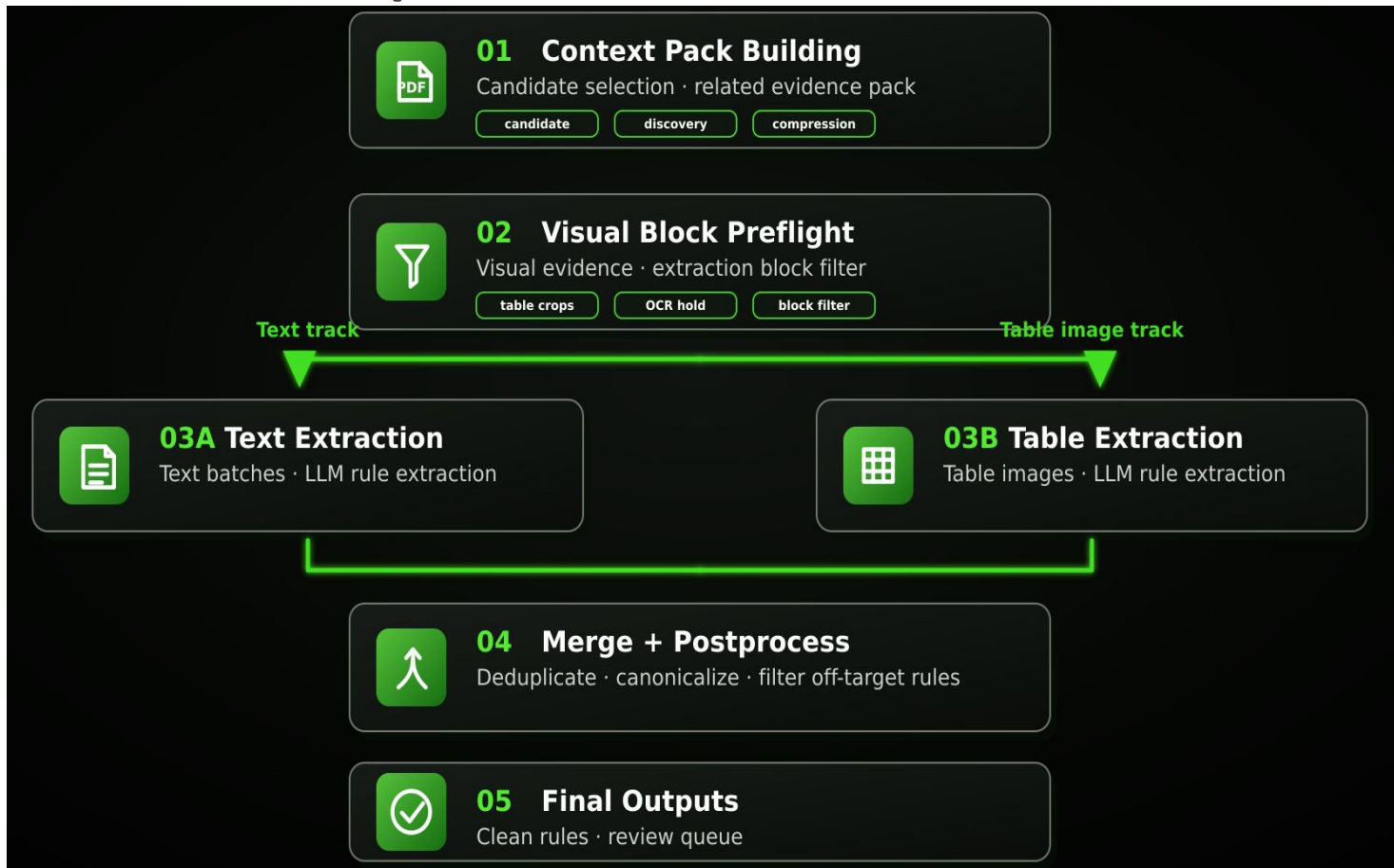
Rule Extraction Challenges

4

Large PDF file with 1053 pages
and 6000+ blocks (Calgary case)



Rule Extraction Pipeline



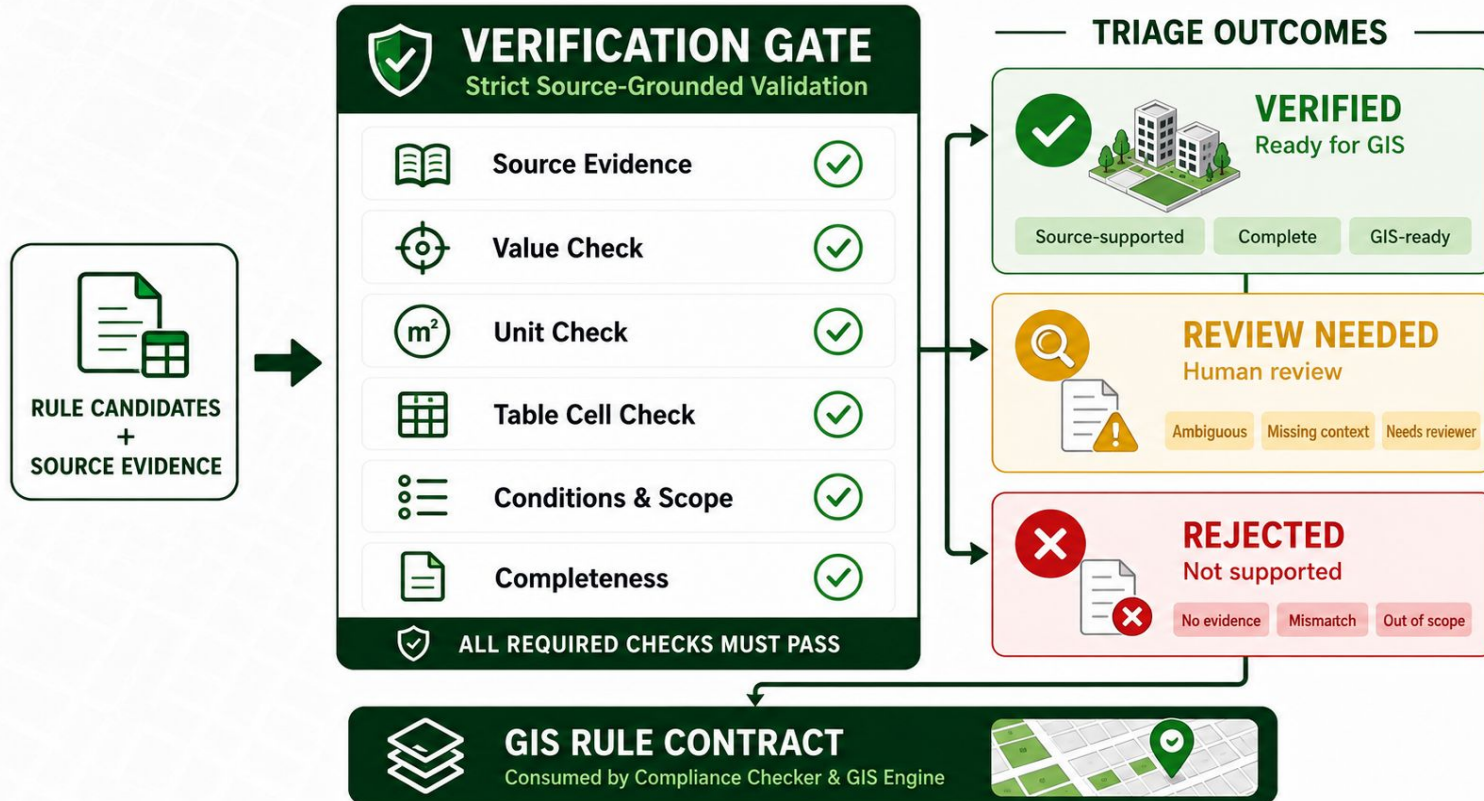
Result Demonstration

Field	Value
Rule object	parking_stall_depth
Constraint	minimum_dimension
Subject	stall depth perpendicular to aisle
Requirement	>= 5.40 m
Condition	Parking angle: 90 degrees
Source	Table 2, page_0195__table_001
Rule ID	pipeline10_rule_0002
Status	kept

Stage 2: Verification Layer

Validating Extracted Rules Against Source Documents

From Extracted AI candidates to Verification layer



M7 Safety Check

AI suggests rules. Verification proves them. GIS exports only verified rules.



PASSED

0 false approvals

Precision

1.00

False verified

0

False approvals

0

Source failures

0

1 Extract

candidate rules



2 Verify

source proof required



3 Export

verified rules only

Per-city outcome

■ Verified ■ Review ■ Rejected ■ Not used

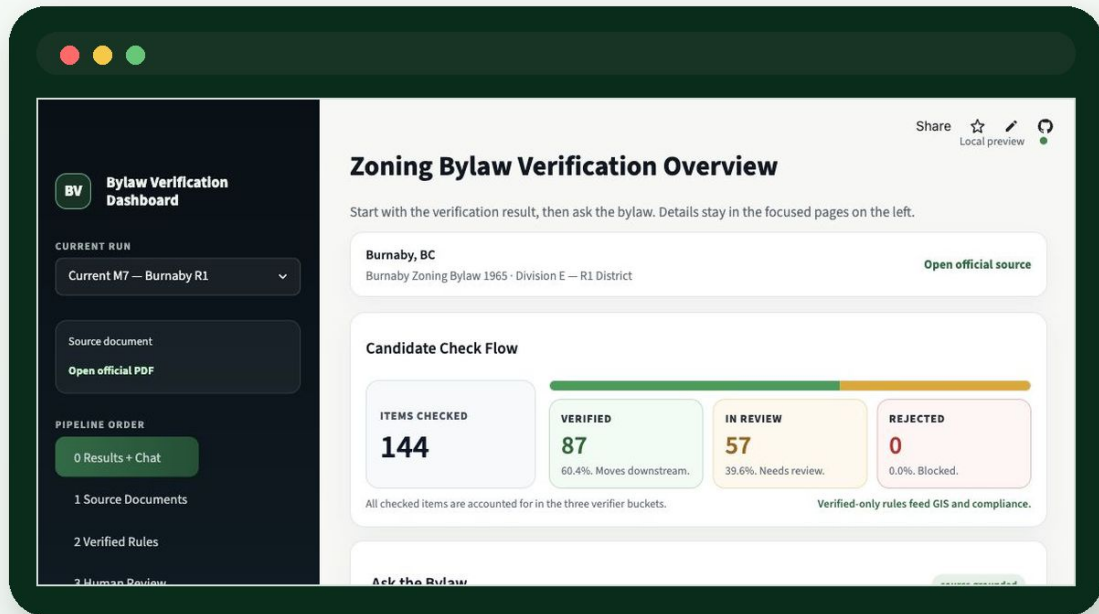
City	Candidates	Split	Precision
Burnaby R1	144	87 57	1.00
Calgary R-CG	150	29 42 27 52	1.00
Vancouver RS	56	10 31 15	1.00

Uncertain rule -> review. Uncertain proposal -> needs_review.

Reviewer Dashboard + QR Access

Updated cloud dashboard

Partner-facing view for verified status, human review, repair evidence, and source-grounded Ask the Bylaw chat.



Scan to open



Streamlit Cloud dashboard

yusenrong46-afk-burnaby-verificat-dashboard

streamlit-app-rcvryj.streamlit.app

Presentation use
scan during demo

Verified status

Human review

Repair evidence

Ask the Bylaw

Read-only access

<https://yusenrong46-afk-burnaby-verificat-dashboardstreamlit-app-rcvryj.streamlit.app/>

Stage 3: Geospatial Rule Verification

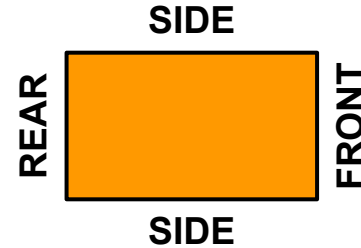
Buildable Area



Given a set of setback distance rules, how much space is left to build on?

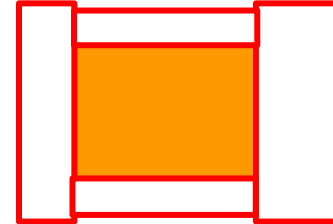
1

Label each boundary as front, side, or rear



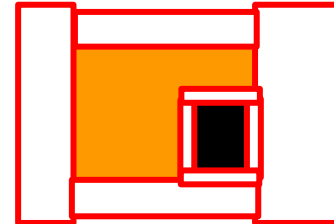
2

Apply setbacks based on labels

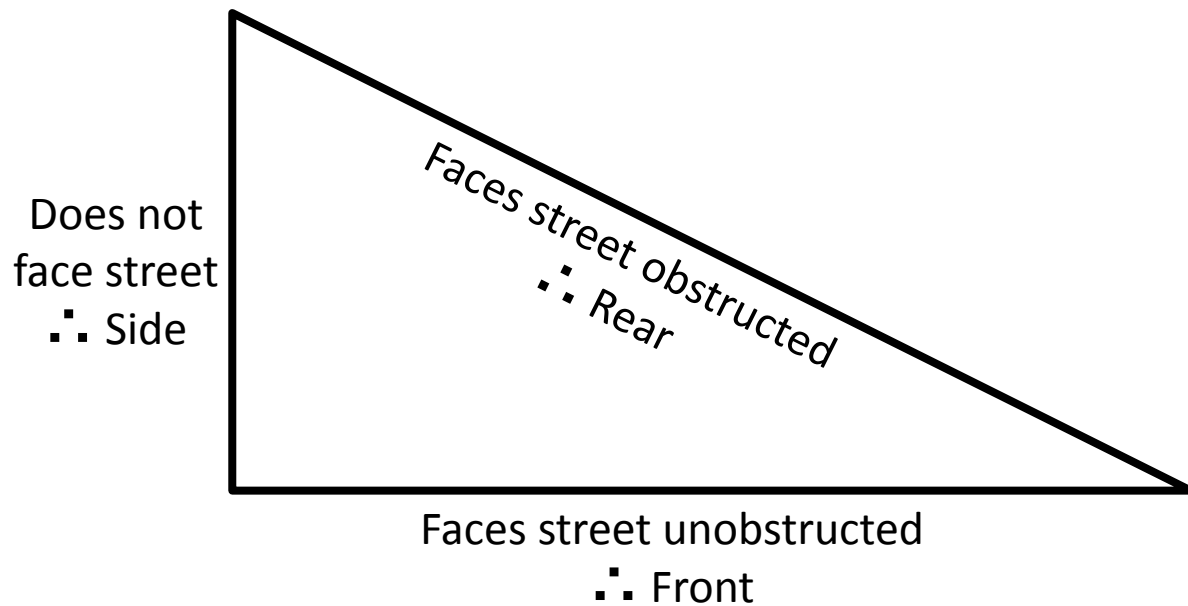


3

Account for existing structures

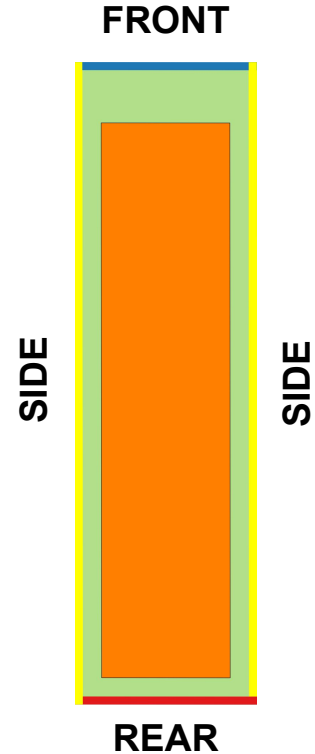


Labelling Boundaries



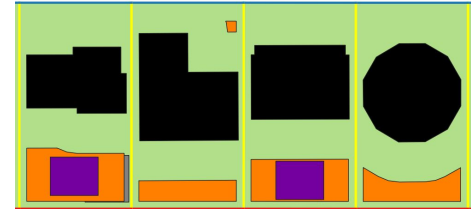
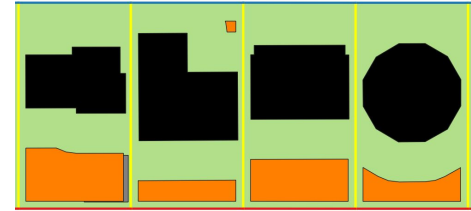
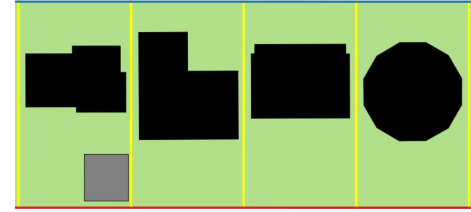
Applying Setbacks

Label	Borders	Setback
Front	Street	Street
Side	Lot	Interior Side
	Street	Street Flanking
	Lane	Interior Side
Rear	Lot	Interior Rear
	Street	Street Flanking
	Lane	Lane



Subtracting Existing Structures

- 1 Label largest building on each lot as principal
- 2 Subtract principal (with buffer) from buildable area
- 3 Fit building footprint



Stage 4: Map Dashboard

Interactive Map: See Every Rule, On Every Lot

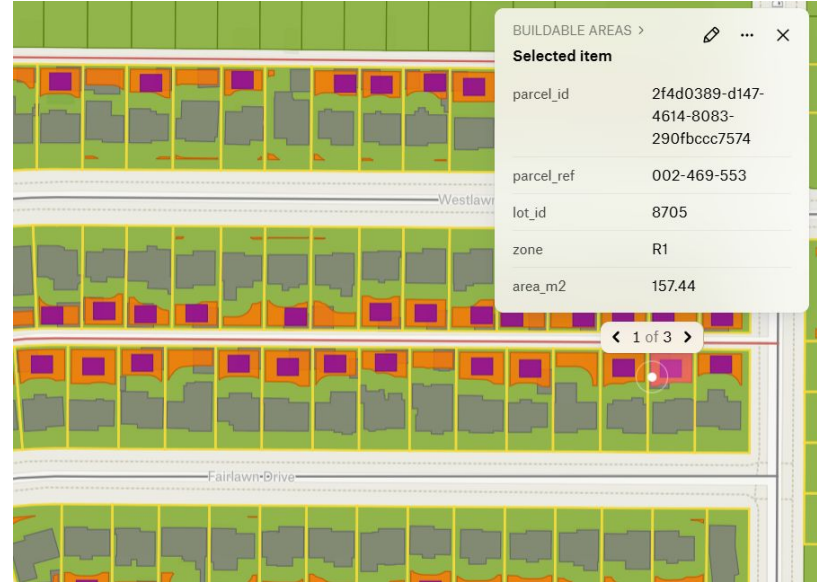
Burnaby



Calgary



Turns a question that used to take an hour into a ten-second answer



Conclusion

Built to scale → one pipeline, any city

Our prototype proves the approach works and can scale up it in the future

Rule Extraction



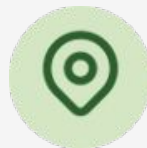
Generalise to extract rules from **all** kinds of bylaw documents

Verification Layer



Check every candidate rule against bylaws evidence

GIS



- Make a “test city”
- Standardize schema for all municipalities

Map Dashboard



Connect all outputs into interactive map